Practical No. 9: Firewall Configuration and Rule-Based Filtering

Aim

To configure and test **inbound and outbound firewall rules** to control network traffic and protect network resources from unauthorized access.

Objectives

- To understand the concept of a firewall.
- To understand inbound and outbound traffic.
- To configure firewall rules based on protocol and port.
- To test the effect of firewall rules.

Theory

Firewall

- A **firewall** is a security mechanism that monitors and controls incoming and outgoing network traffic based on predefined rules.

Inbound Rule

- An **inbound rule** controls traffic **entering** the computer.
- **Example:** Block incoming TCP traffic on port 80.

Outbound Rule

- An **outbound rule** controls traffic **leaving** the computer.
- **Example:** Block outgoing TCP connections to remote port 80.

Rule-Based Filtering

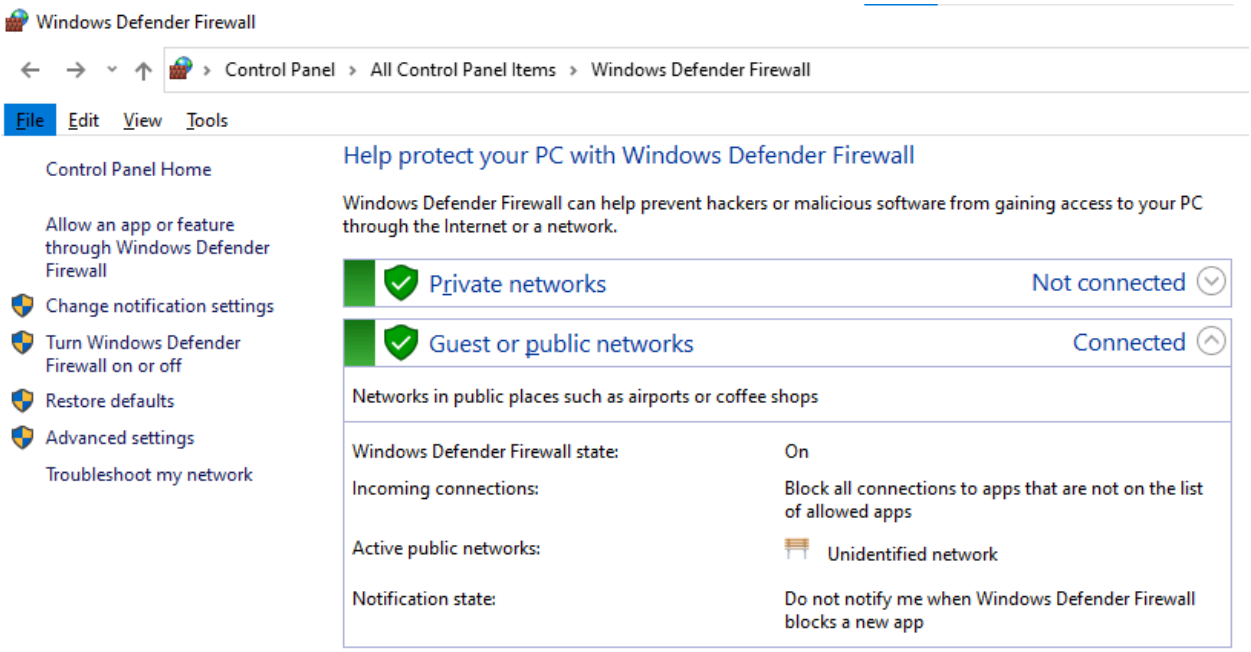
Firewall rules can filter traffic based on:

- Protocol – TCP/UDP
- Port number
- IP address
- Direction
- Network profile

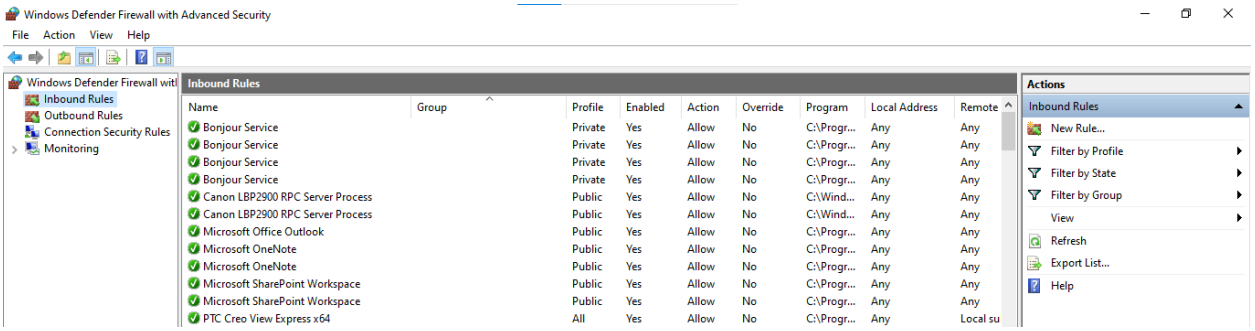
Procedure

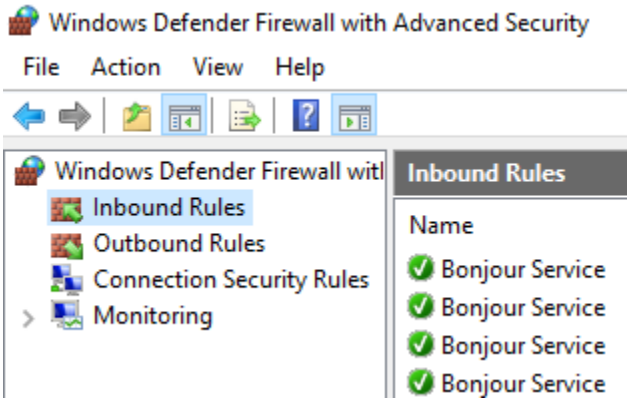
A. Configure Inbound Rule

1. **Start → Control Panel → Windows firewall → Advanced settings →**

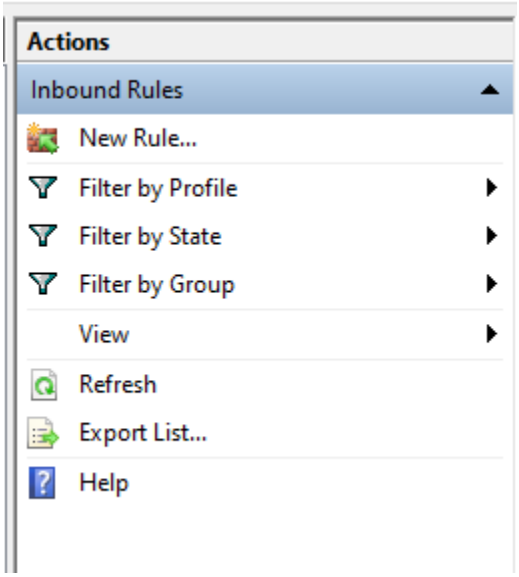


2. Left hand side of the window shows a list of rules. From the list, select **Inbound Rules.**

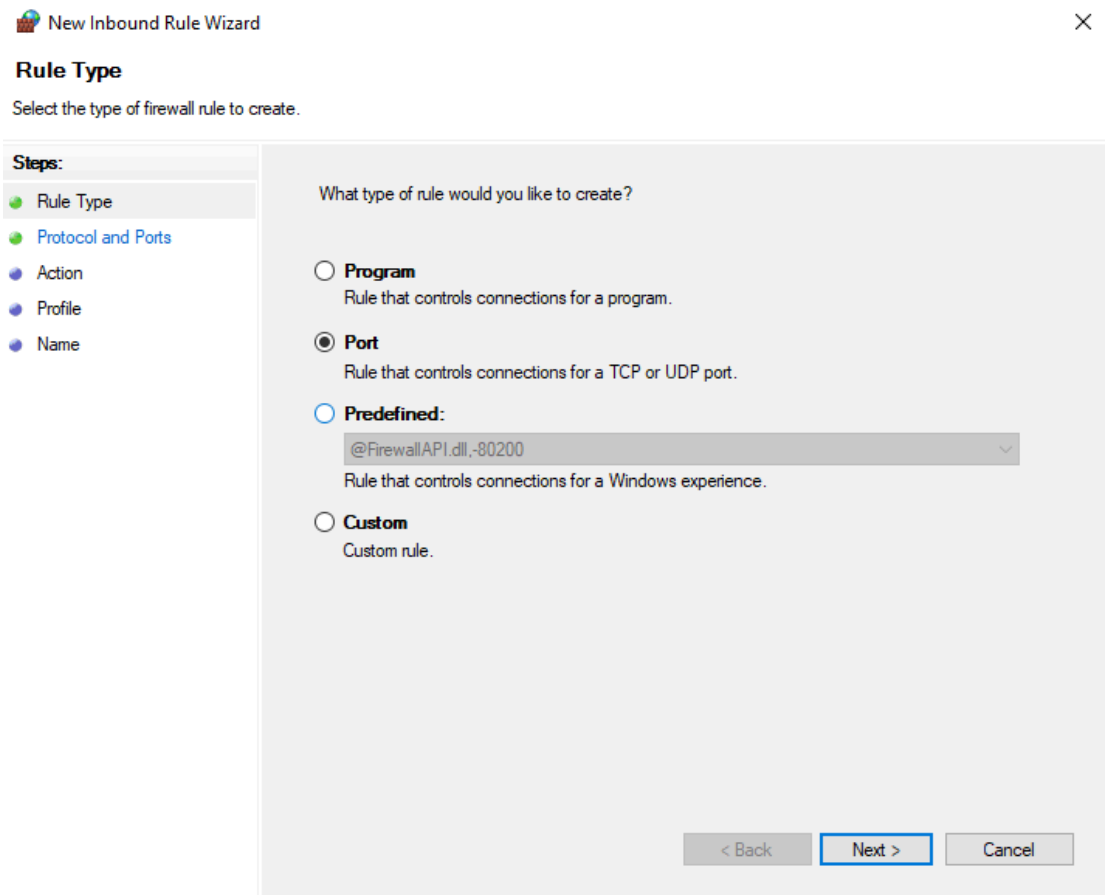




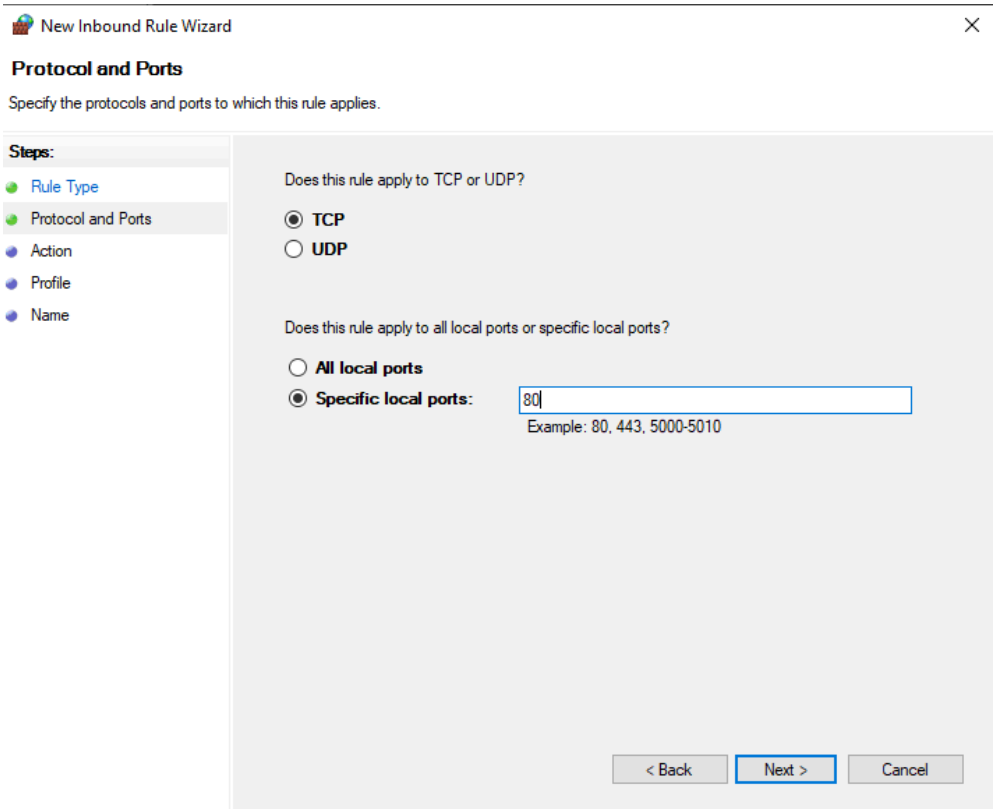
3. Then, from right pane select option **"New Rule"**



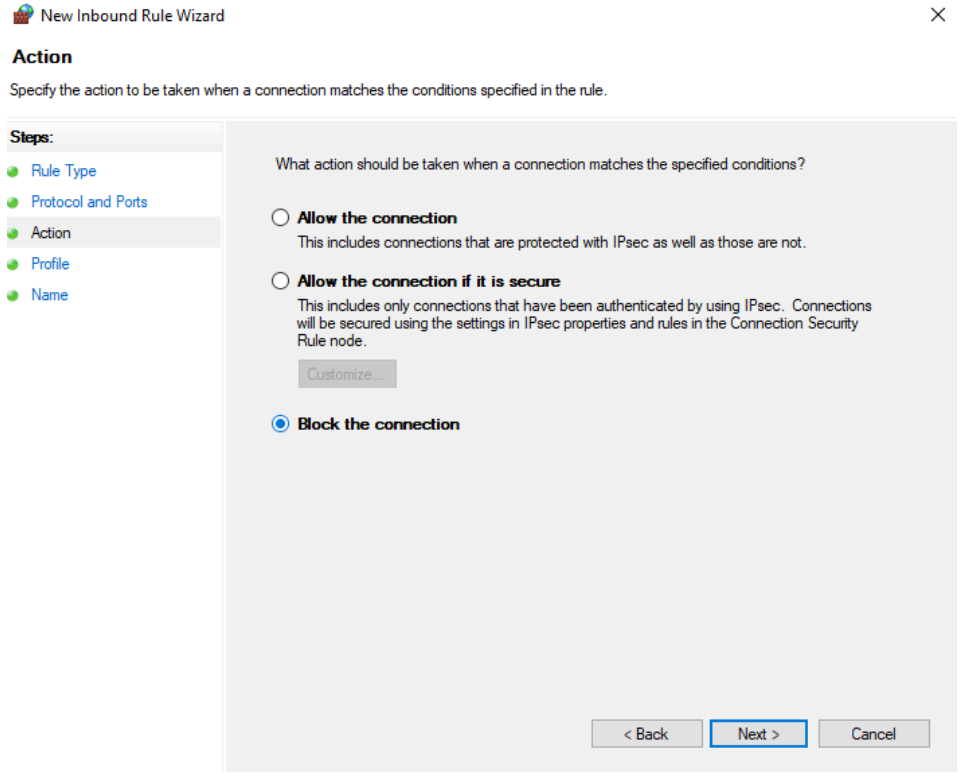
4. It will open the '**New Inbound Rule Wizard**' window.
5. From it, select '**Port'** as the new Rule Type and click Next.



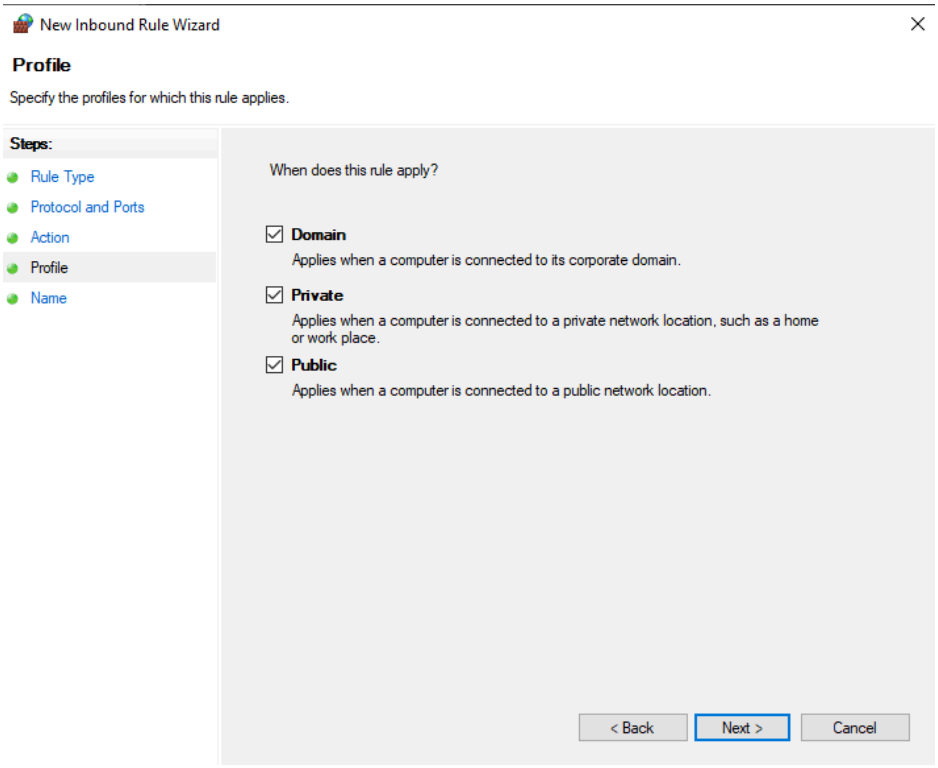
6. Select the option "**TCP**" and in the field '**Specific local ports**' enter **'80'**.
7. Click Next to continue.



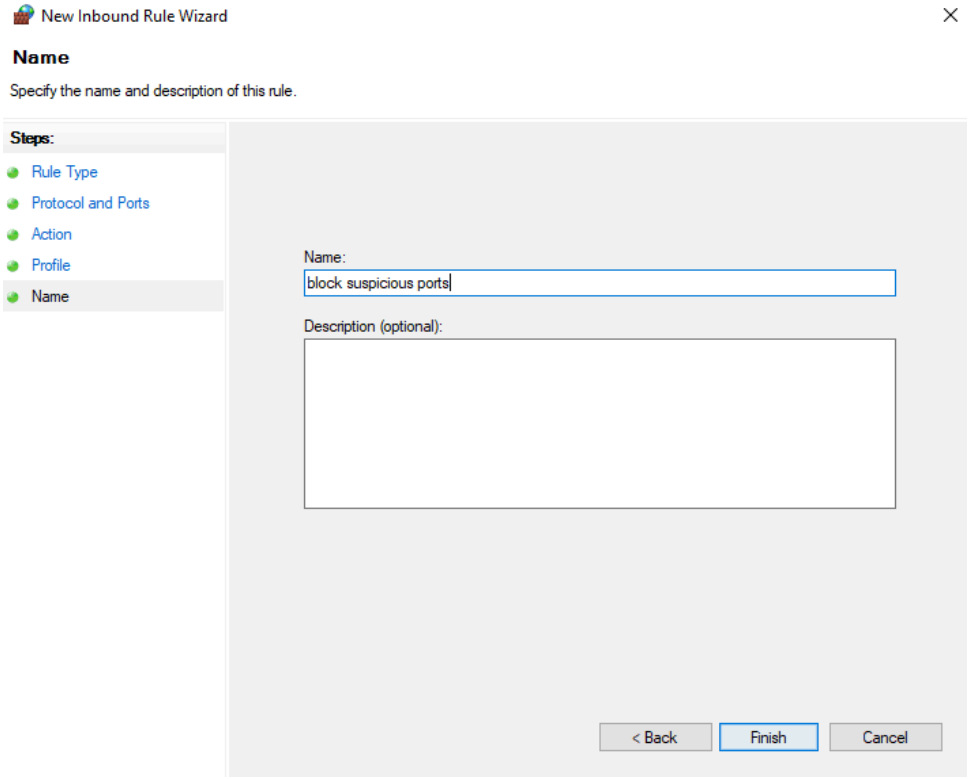
8. Next, select '**Block the connection**' as the Action and click **Next**.



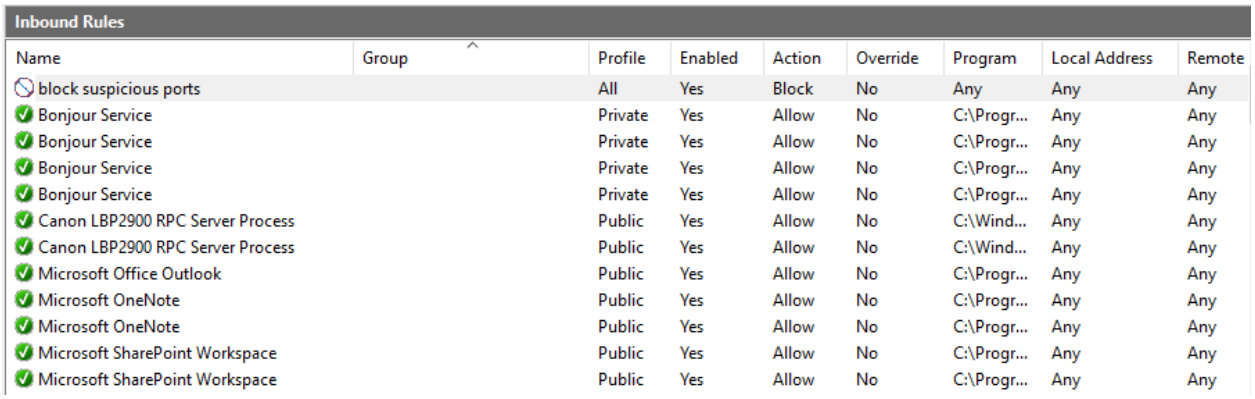
9. Later, select all the profiles available for different types of connections (**Domain, Private and Public**) and Click **Next** to continue.



10. Enter the name of your choice to the new rule '**block suspicious ports**'. Adding description for the same rule is optional.



11. Finally, click the **Finish** button to configure settings.

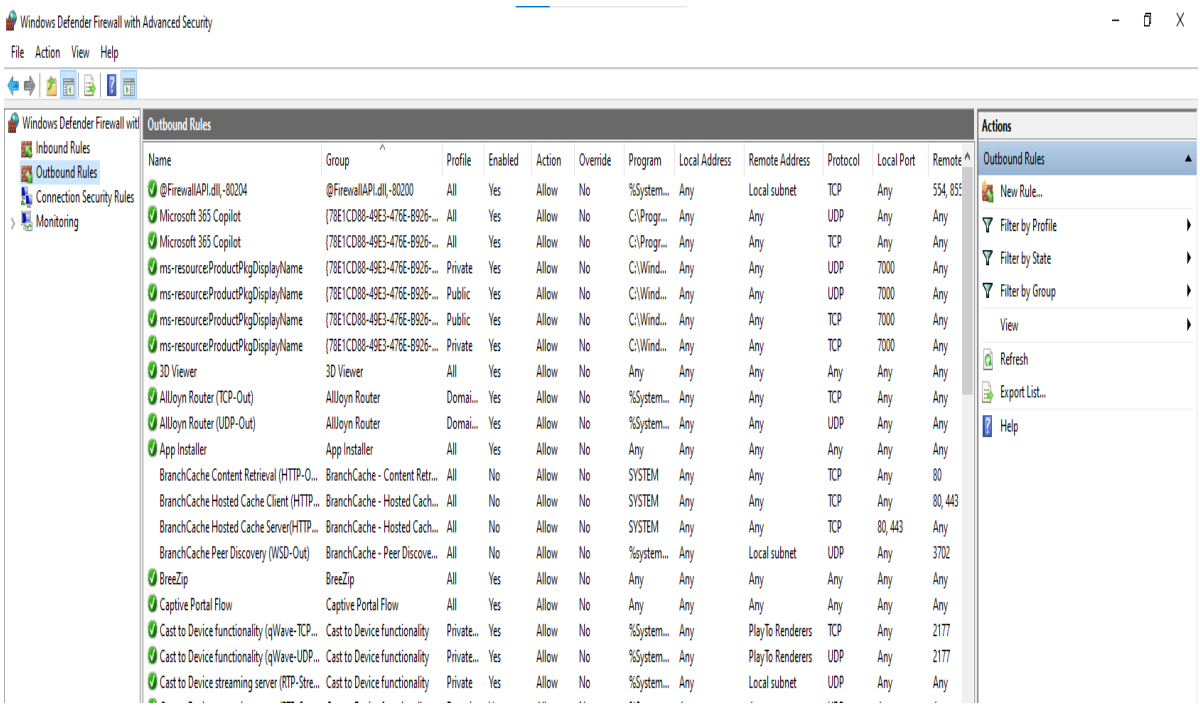


Inbound Rules

Name	Group	Profile	Enabled	Action	Override	Program	Local Address	Remote
block suspicious ports		All	Yes	Block	No	Any	Any	Any
Bonjour Service		Private	Yes	Allow	No	C:\Progr...	Any	Any
Bonjour Service		Private	Yes	Allow	No	C:\Progr...	Any	Any
Bonjour Service		Private	Yes	Allow	No	C:\Progr...	Any	Any
Bonjour Service		Private	Yes	Allow	No	C:\Progr...	Any	Any
Canon LBP2900 RPC Server Process		Public	Yes	Allow	No	C:\Wind...	Any	Any
Canon LBP2900 RPC Server Process		Public	Yes	Allow	No	C:\Wind...	Any	Any
Microsoft Office Outlook		Public	Yes	Allow	No	C:\Progr...	Any	Any
Microsoft OneNote		Public	Yes	Allow	No	C:\Progr...	Any	Any
Microsoft OneNote		Public	Yes	Allow	No	C:\Progr...	Any	Any
Microsoft SharePoint Workspace		Public	Yes	Allow	No	C:\Progr...	Any	Any
Microsoft SharePoint Workspace		Public	Yes	Allow	No	C:\Progr...	Any	Any

B. Configure Outbound Rule

1. In **Windows Defender Firewall with Advanced Security**, select **Outbound Rules**.



Windows Defender Firewall with Advanced Security

File Action View Help

Windows Defender Firewall with...
- Inbound Rules
- Outbound Rules
- Connection Security Rules
- Monitoring

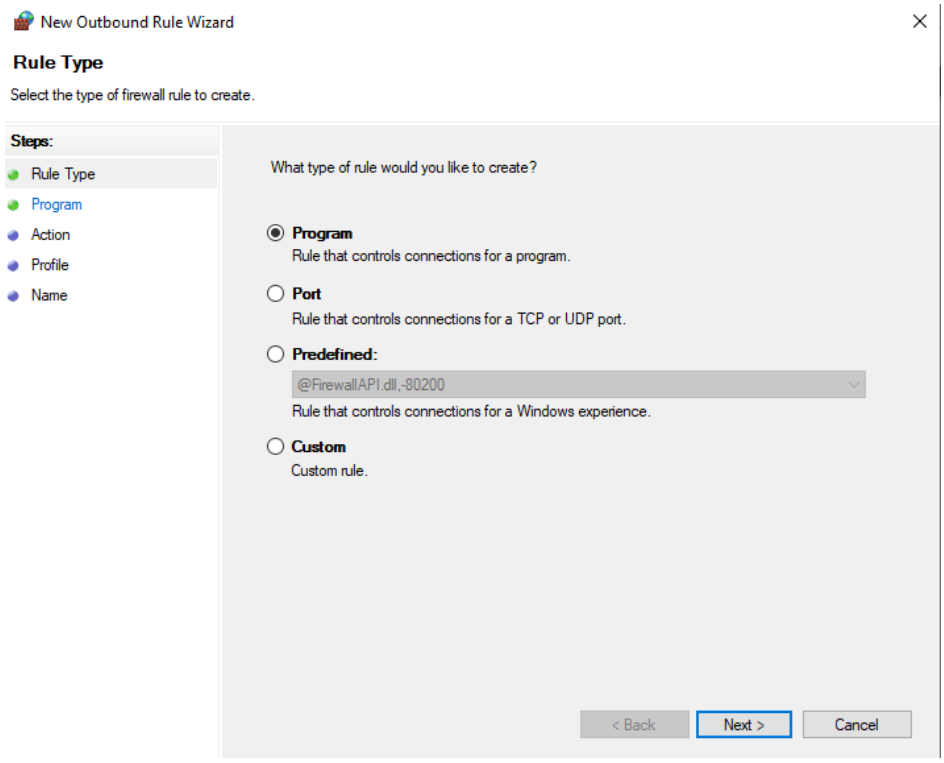
Outbound Rules

Name	Group	Profile	Enabled	Action	Override	Program	Local Address	Remote Address	Protocol	Local Port	Remote
@FirewallAPI.dll,-80204	@FirewallAPI.dll,-80200	All	Yes	Allow	No	%System...	Any	Local subnet	TCP	Any	554, 855
Microsoft 365 Copilot	{78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progr...	Any	Any	UDP	Any	Any
Microsoft 365 Copilot	{78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progr...	Any	Any	TCP	Any	Any
ms-resourceProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...	Private	Yes	Allow	No	C:\Wind...	Any	Any	UDP	7000	Any
ms-resourceProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...	Public	Yes	Allow	No	C:\Wind...	Any	Any	UDP	7000	Any
ms-resourceProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...	Public	Yes	Allow	No	C:\Wind...	Any	Any	TCP	7000	Any
ms-resourceProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...	Private	Yes	Allow	No	C:\Wind...	Any	Any	TCP	7000	Any
3D Viewer	3D Viewer	All	Yes	Allow	No	Any	Any	Any	Any	Any	Any
AllJoyn Router (TCP-Out)	AllJoyn Router	Domai...	Yes	Allow	No	%System...	Any	Any	TCP	Any	Any
AllJoyn Router (UDP-Out)	AllJoyn Router	Domai...	Yes	Allow	No	%System...	Any	Any	UDP	Any	Any
App Installer	App Installer	All	Yes	Allow	No	Any	Any	Any	Any	Any	Any
BranchCache Content Retrieval (HTTP-O...	BranchCache - Content Retr...	All	No	Allow	No	SYSTEM	Any	Any	TCP	Any	80
BranchCache Hosted Cache Client (HTTP...	BranchCache - Hosted Cach...	All	No	Allow	No	SYSTEM	Any	Any	TCP	Any	80, 443
BranchCache Hosted Cache Server(HTTP...	BranchCache - Hosted Cach...	All	No	Allow	No	SYSTEM	Any	Any	TCP	80, 443	Any
BranchCache Peer Discovery (WSD-Out)	BranchCache - Peer Discove...	All	No	Allow	No	%system...	Any	Local subnet	UDP	Any	3702
BreeZip	BreeZip	All	Yes	Allow	No	Any	Any	Any	Any	Any	Any
Captive Portal Flow	Captive Portal Flow	All	Yes	Allow	No	Any	Any	Any	Any	Any	Any
Cast to Device functionality (qWave-TCP...	Cast to Device functionality	Private...	Yes	Allow	No	%System...	Any	PlayTo Renderers	TCP	Any	2177
Cast to Device functionality (qWave-UDP...	Cast to Device functionality	Private...	Yes	Allow	No	%System...	Any	PlayTo Renderers	UDP	Any	2177
Cast to Device streaming server (RTP-Stre...	Cast to Device functionality	Private	Yes	Allow	No	%System...	Any	Local subnet	UDP	Any	Any

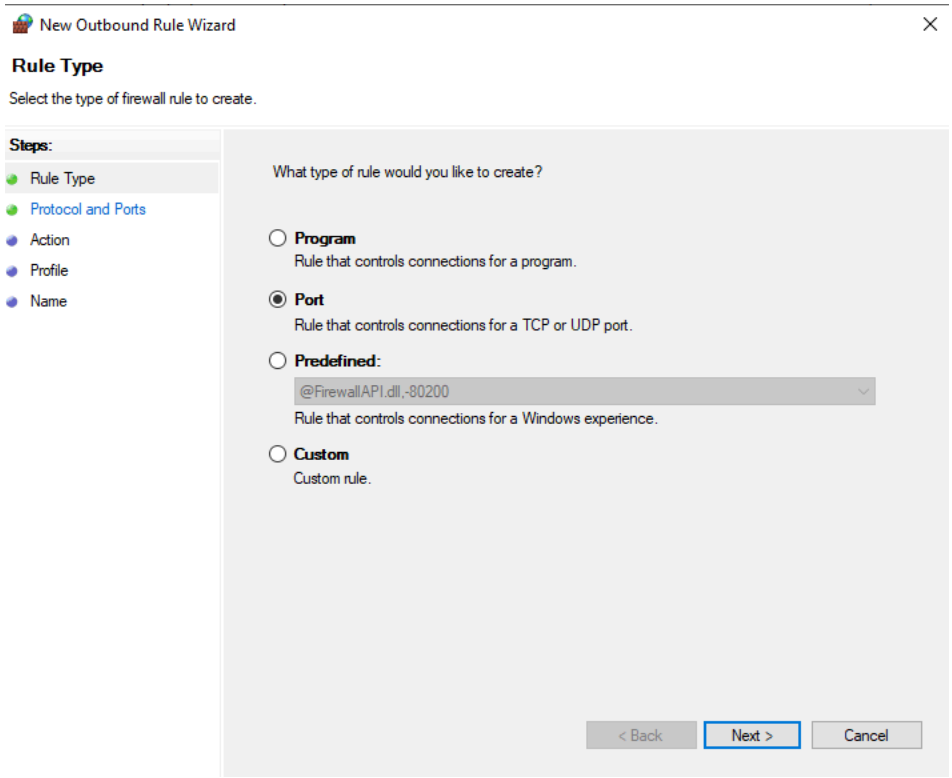
Actions

Outbound Rules
- New Rule...
- Filter by Profile
- Filter by State
- Filter by Group
- View
- Refresh
- Export List...
- Help

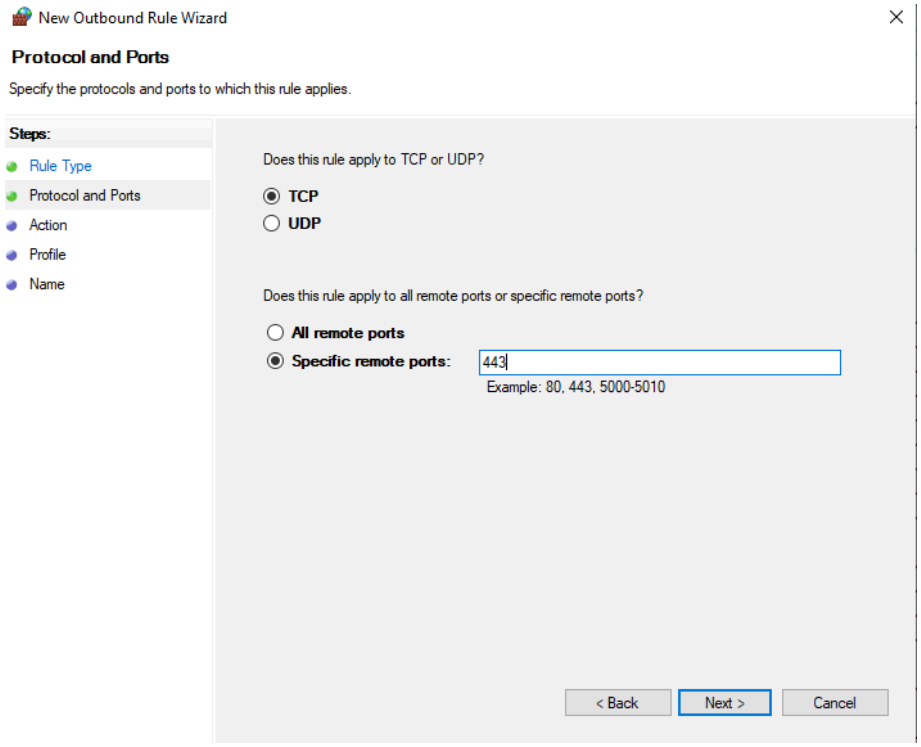
2. Click **New Rule**.



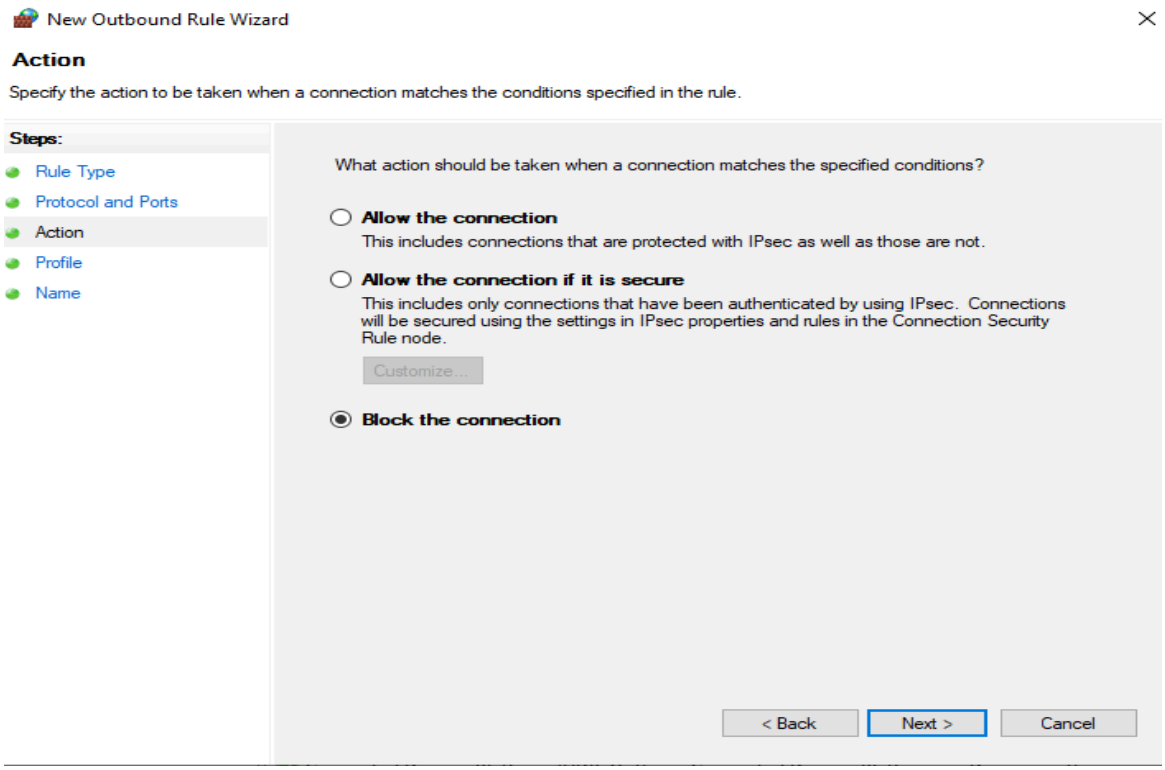
3. Select **Port → Next**.



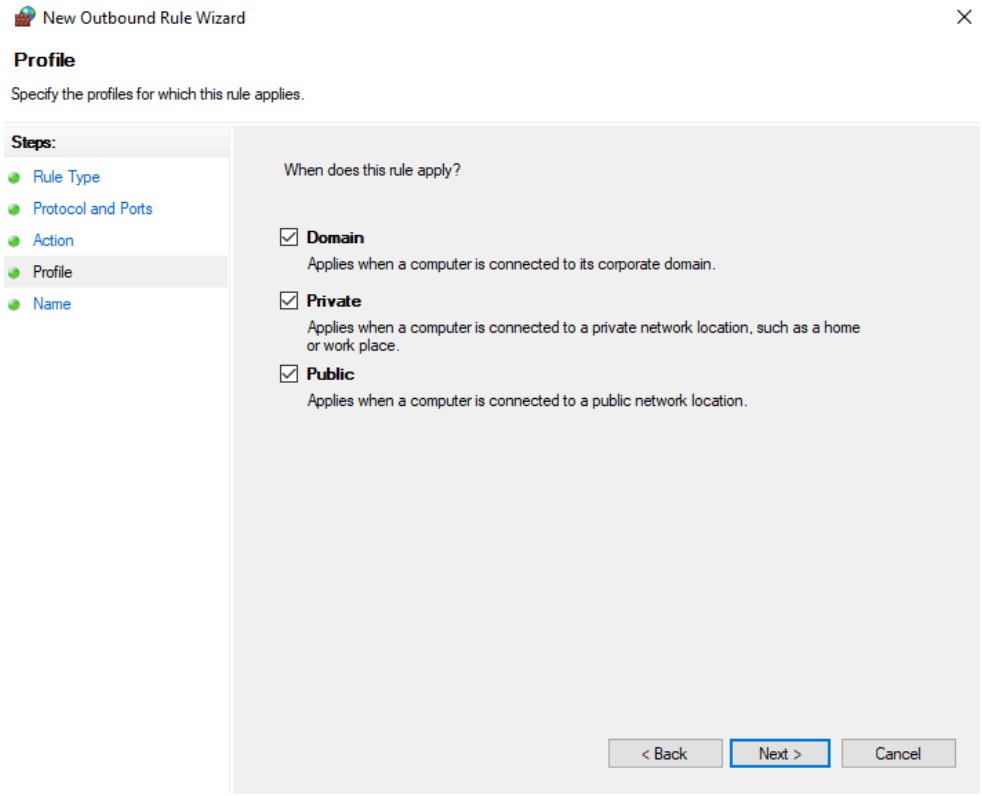
4. Select **TCP** and enter **443** under **Specific remote ports**.



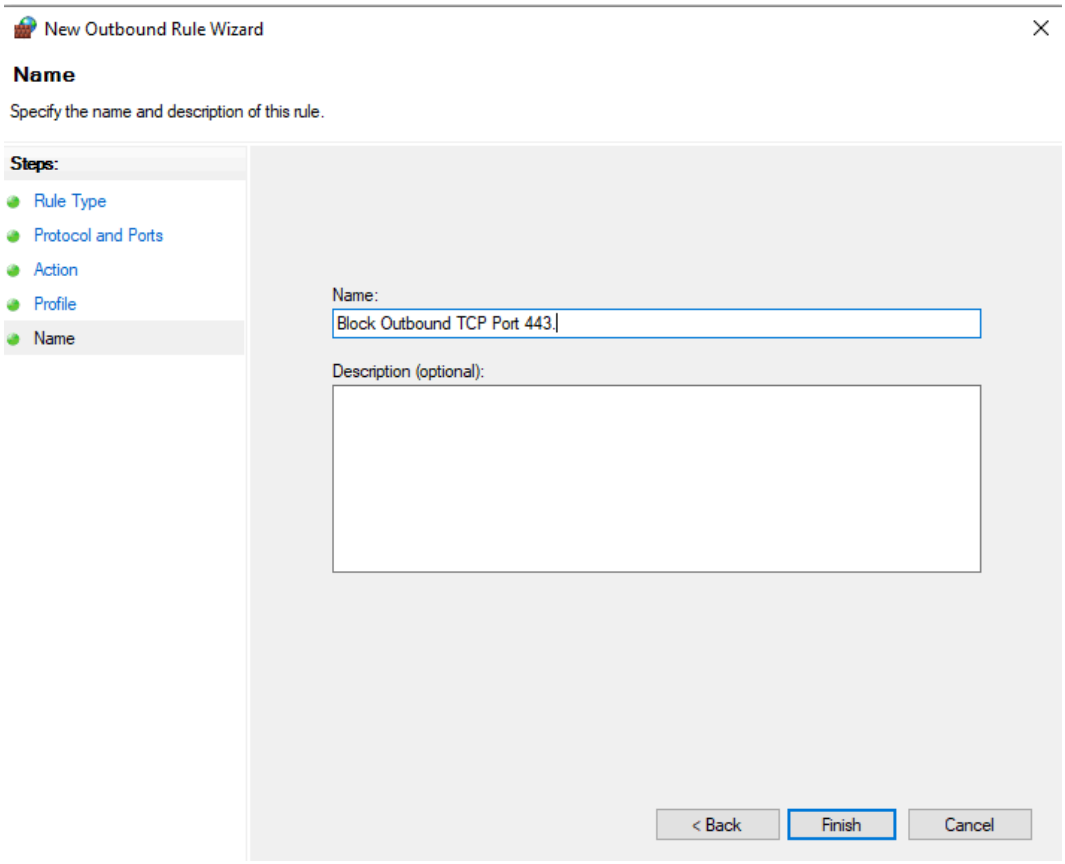
5. Select **Block the connection → Next**.



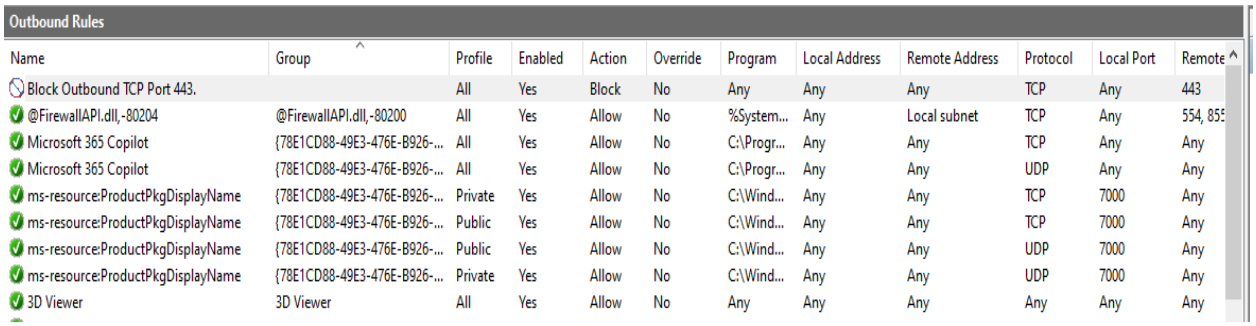
6. Select **Domain, Private and Public** profiles → **Next**.



7. Enter rule name: **Block Outbound TCP Port 443**.



8. Click **Finish**.



Name	Group	Profile	Enabled	Action	Override	Program	Local Address	Remote Address	Protocol	Local Port	Remote
Block Outbound TCP Port 443.		All	Yes	Block	No	Any	Any	Any	TCP	Any	443
@FirewallAPI.dll,-80204	@FirewallAPI.dll,-80200	All	Yes	Allow	No	%System...	Any	Local subnet	TCP	Any	554, 855
Microsoft 365 Copilot	{78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progr...	Any	Any	TCP	Any	Any
Microsoft 365 Copilot	{78E1CD88-49E3-476E-B926-...	All	Yes	Allow	No	C:\Progr...	Any	Any	UDP	Any	Any
ms-resource:ProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...	Private	Yes	Allow	No	C:\Wind...	Any	Any	TCP	7000	Any
ms-resource:ProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...	Public	Yes	Allow	No	C:\Wind...	Any	Any	TCP	7000	Any
ms-resource:ProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...	Public	Yes	Allow	No	C:\Wind...	Any	Any	UDP	7000	Any
ms-resource:ProductPkgDisplayName	{78E1CD88-49E3-476E-B926-...	Private	Yes	Allow	No	C:\Wind...	Any	Any	UDP	7000	Any
3D Viewer	3D Viewer	All	Yes	Allow	No	Any	Any	Any	Any	Any	Any